GRADE 11 · TERM 4

The Connected World & Exam Readiness

A pencil-and-paper workbook: understand the internet of things and big data, then get ready for the exams and the PAT.

Name: _____

Class: _____ Date: _____

How to use this workbook

1. Work with a **pencil**. Mistakes are how we learn: rub out and try again.
2. Start each unit with **Think First**. Try it on your own before you read the explanation.
3. Read **Learn It**, then do the activities. Trace tables and code are written **by hand**.
4. At the end of each unit, tick the "I can" boxes you are sure of. Write down anything you still want to ask.
5. The **checkpoint test** is done under test conditions, with no notes.

In this workbook

1. The connected world
2. Media and the web
3. Your digital footprint
4. Revision workout: practical part 1
5. Revision workout: practical part 2
6. Revision workout: theory on paper
7. PAT checklist and exam skills
8. Term checkpoint test

> **Matches the Annual Teaching Plan:** Term 4 — PAT finalisation; revision of all Term 1–3 practical work; evolution of the internet (IoT, 4IR); Big Data (volume, variety, velocity); download vs streaming, live broadcasts, VOD, IPTV; compression (lossy and lossless); static vs dynamic websites; HTTP vs HTTPS; MFA, OTP and security tokens; location-based services; internet careers; social implications of Big Data, IoT and globalisation; protecting your online identity.

UNIT 1 The connected world

Your phone, the traffic light outside school and the electricity meter at home are starting to talk to computers on the internet. You will learn what the Internet of Things is, why it creates so much data, and how this wave of technology is changing jobs and society.

By the end of this unit I can: (tick each box when you are sure)

- ☐ describe how the web has changed, up to the Internet of Things
- ☐ list the characteristics and uses of IoT devices
- ☐ explain volume, velocity and variety with examples
- ☐ work out how much data sensors produce
- ☐ write a paragraph on how 4IR affects jobs

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

1.1 Count the connected things

Write down **six** things at home, at school or on the road to school that contain a sensor or small computer and are (or could be) connected to the internet. For **two** of them say what the sensor measures, where the data goes and what decision could be made *without a person*.

LEARN IT

From web pages to the Internet of Things

Web 1.0 (1990s) pages were mostly read-only. **Web 2.0** is the read-write web of blogs, wikis and social media, where users create the content. **Web 3.0** is the emerging "intelligent" web that tries to understand the meaning of data. The internet has also grown from connecting *people* to connecting *things*.

The **Internet of Things (IoT)** is a network of physical objects that have **sensors**, a small processor and an internet connection. Its characteristics: each device has its own network address; it senses something in the real world; it sends and receives data **automatically**, without a person typing; and it can often be monitored or controlled from far away. Uses: smart homes, wearables, remote health monitoring, soil sensors on farms, connected cars and smart cities. Risks: privacy, hacking of unprotected devices, and dependence on electricity and connectivity (load shedding switches many of them off).

LEARN IT

Big Data and the Fourth Industrial Revolution

Big Data is data that is too large, too fast or too mixed for an ordinary spreadsheet or single database to handle. Think of Mama Dlamini's exercise book against a supermarket chain that records every barcode scan, card payment, truck position and customer post. The three Vs describe the difference.

V	Meaning	Example
Volume	The *amount* of data, from terabytes upward	A mobile network stores billions of call records
Velocity	The *speed* at which data arrives and must be processed	A bank checks each card swipe for fraud in a fraction of a second
Variety	Many *forms*: tables, text, images, video, GPS and sensor data	A hospital keeps X-rays, doctors' notes and heart-rate readings

Industrial revolutions: 1st steam and water power; 2nd electricity and assembly lines; 3rd electronics, computers and automation; **4th (4IR)** merges the physical, digital and biological worlds through AI, robotics, IoT, Big Data and 3D printing. **Employment:** routine jobs (till operators, assembly work) can be automated and disappear, while new jobs such as data analyst and robot technician appear, so workers must retrain. **Society:** life becomes more convenient and healthcare and education reach remote places, but people without devices or data are left behind (the **digital divide**) and more data about everyone is collected, which threatens privacy.

1.2 Key words [8]

In Web 1.0 users could only _____ pages; Web 2.0 lets users _____ content. A network of objects with sensors that exchange data automatically is the _____. The three Vs of Big Data are _____, _____ and _____. The 4IR merges the _____, digital and biological worlds. The gap between people who have technology and those who do not is the _____.

1.3 Which V? [4]

Circle the V of Big Data that each situation shows best.

1	A bank must check each card swipe for fraud within a fraction of a second.	**Volume**	**Velocity**	**Variety**
2	A mobile network stores billions of call records.	**Volume**	**Velocity**	**Variety**
3	A hospital keeps X-ray pictures, doctors' notes and tables of lab results.	**Volume**	**Velocity**	**Variety**
4	A news site must show trending topics from posts *as they are written*.	**Volume**	**Velocity**	**Variety**

1.4 How much data? (Volume) [5]

A weather sensor sends **200 bytes** every **30 seconds**, all day and night. Work out the data from **500** sensors in one day (1 KB = 1 024 bytes; 1 MB = 1 024 KB).

Step	Answer
1. Readings per sensor per day (24 h × 60 min × 2 per minute)	
2. Bytes per sensor per day (readings × 200)	
3. KB per sensor per day (÷ 1 024)	
4. KB for 500 sensors (× 500)	
5. MB for 500 sensors (÷ 1 024, 2 decimal places)	

1.5 Discuss: smart water meters [4]

A municipality fits **smart water meters** to 5 000 homes; each sends a reading over the internet every hour. In a short paragraph discuss **one benefit** and **one risk** of this IoT system.

1.6 4IR at the checkout [4]

Supermarkets are adding self-checkout tills and shelf-stocking robots. In a paragraph explain how this may change **employment**: name a job that may **disappear**, one that may be **created**, and say what workers must do.

How did I do with this unit? ☐ I got it ☐ Nearly there ☐ I need help

One thing I want to ask my teacher:

UNIT 2 Media and the web

Every video you stream and photo you send has been squashed to travel faster. This unit shows how the squashing works, how websites are built and how a site can prove it is safe, with all calculations done by hand.

By the end of this unit I can: (tick each box when you are sure)

- ☐ compare downloading and streaming; explain live broadcasts, VOD and IPTV
- ☐ explain lossy and lossless compression
- ☐ calculate a file size and the data used by a video stream
- ☐ compare static and dynamic websites
- ☐ explain HTTPS, MFA, OTPs and security tokens

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

2.1 Squash the message

Sipho's rule: write **how many times** a letter repeats, then the letter. So **AAAABBBCCDAA** becomes **4A3B2C1D2A**. (a) Squash **WWWWWWBBW**. (b) Unsquash **3Q2Z1K**. (c) Can you always rebuild the exact original? A clerk saves space by rounding each mark to the nearest 10. She stores **67, 72, 84, 91, 58** as ____ , ____ , ____ , ____ , ____ . (d) Can she rebuild the originals? Which rule loses information?

LEARN IT

Download, stream, broadcast, VOD and IPTV

Sipho downloads a match overnight; Lerato streams it, and load shedding switches her router off halfway. Only Sipho can keep watching. When you **download**, the whole file is copied to your device, so you keep it and can use it offline. When you **stream**, data is played as it arrives and no copy is kept; a few seconds are held in a **buffer** so playback does not stutter, and it needs a steady connection and a lot of bandwidth. A **live broadcast** is streamed as it happens (a match, a news bulletin) so you cannot skip ahead. **Video on demand (VOD)** lets you choose what to watch and when from a library on servers (Netflix, Showmax). **IPTV** delivers TV channels over an internet connection instead of an aerial or dish, with pause, rewind and catch-up.

LEARN IT

Compression

Compression makes a file smaller so it uses less space and travels faster. **Lossless** compression keeps every bit, so the original can be rebuilt exactly (ZIP, PNG). **Lossy** compression permanently throws away detail people are unlikely to notice, so the file is much smaller but the original cannot be rebuilt (MP3 audio, JPEG photos, MPEG-4 video). More compression means a smaller file but lower quality.
Picture size = width × height (pixels) × 3 bytes per pixel (true colour). **Saving** (%) = (original − compressed) ÷ original × 100. A ratio of 15:1 means the compressed file is one fifteenth of the original.

LEARN IT

Websites and secure logins

A **static** website shows the same fixed pages to everyone and changes only when the developer edits the files; it is quick and cheap. A **dynamic** website builds each page when asked, usually from a database, so it can show your name, cart or today's prices. **HTTP** sends pages as plain text that anyone intercepting them can read; **HTTPS** encrypts the data (SSL/TLS) and shows a padlock, so any page that takes a password or card number must use it.

Multi-factor authentication (MFA) asks for proofs from two different groups: something you **know** (password), **have** (phone, token) or **are** (fingerprint). A **one-time PIN (OTP)** works once and only for a few minutes. A **security token** (a device or app) shows a new code every 30 to 60 seconds. A stolen password alone is then useless, although SMS codes can be intercepted and a lost phone is a problem.

2.2 Lossy or lossless? [5]

Circle the type of compression that fits each item.

1	An MP3 song on a phone	**Lossy** **Lossless**
2	A ZIP file of school documents	**Lossy** **Lossless**
3	A PNG logo that must stay perfectly sharp	**Lossy** **Lossless**
4	A hospital X-ray that may lose no detail	**Lossy** **Lossless**
5	An MPEG-4 video streamed on a slow connection	**Lossy** **Lossless**

2.3 Squash and unsquash [5]

Use Sipho's rule (count, then letter). Then look at the answer for **ABCD**: is it shorter or longer than the original?

Squash or unsquash	Answer
AAAABBBCCDAA	
XXYYYY	
ABCD	
Unsquash 5Z2Q	
Unsquash 1M1N4O	

2.4 File size by hand [5]

A true-colour photo is **1 200 × 800 pixels**, 3 bytes per pixel. Work out its size and the size after JPEG compression at **15:1** (1 KB = 1 024 bytes).

Working

```
pixels := width * height
bytes  := pixels * 3
KB     := bytes / 1024
MB     := KB / 1024
JPEG   := bytes / 15
```

Step	Working	Answer
1	1 200 × 800	
2	pixels × 3	
3	bytes ÷ 1 024	
4	KB ÷ 1 024 (2 decimals)	
5	bytes ÷ 15, then ÷ 1 024	

2.5 Streaming costs [4]

A stream uses **3 MB per minute**. Data costs **R85 per GB** (1 GB = 1 024 MB).

Question	Answer
(a) Data used by a 90-minute match	
(b) Cost of that data to the nearest cent (270 ÷ 1 024 × R85)	
(c) How many **whole** 90-minute matches fit into a 2 GB bundle?	
(d) Time to download the 270 MB file at 8 megabits per second (1 byte = 8 bits)	

2.6 How long is a code valid? [4]

A bank sends an OTP at **14:02:10**; it is valid for **5 minutes**. Work out the time since sending and whether each code is accepted.

Typed in at	Time since sent	Accepted or rejected?
14:03:05		
14:06:50		
14:07:25		

2.7 Why the S matters **[3]**

Explain why a school's online fee-payment page must use **HTTPS** and not plain HTTP.

2.8 Stolen password, safe account **[3]**

A criminal steals Thandi's online banking password. Explain how **MFA** with an OTP still protects her money.

2.9 Motivate: static or dynamic? **[4]**

A school wants one site with term dates and a contact page, and a second site where parents log in to see their child's fees. **Motivate** which should be static and which dynamic.

How did I do with this unit? ☐ I got it ☐ Nearly there ☐ I need help

One thing I want to ask my teacher:

UNIT 3 Your digital footprint

Everything you post, search and tap leaves a trail. This unit is about who can follow it, what your phone knows about where you have been, how to protect yourself, and which careers and social effects come with the connected world.

By the end of this unit I can: (tick each box when you are sure)

- ☐ tell an active from a passive digital footprint
- ☐ explain location-based services, with benefits and risks
- ☐ protect my online identity
- ☐ match internet careers to their work
- ☐ discuss social effects of Big Data, IoT and globalisation

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

3.1 What does the panel find?

Zinhle applies for a bursary. The panel searches her name and finds: a public photo of her in school uniform at a party, a rude joke from Grade 9, a gold medal result in school sport, and a petition for clean water that she signed. Which results help her and which could harm her? Write **three things** she should do now.

_____

_____

_____

_____

LEARN IT

Online identity, footprint and location

Your **online identity** is the picture of you that others build from your profiles, posts and comments. Your **digital footprint** is the trail of data you leave. An **active** footprint is data you share on purpose (posts, forms); a **passive** footprint is collected without you doing anything (cookies, your IP address, location logs). Content can be copied and stored by others, and employers, universities and criminals look at it. **Protect yourself:** keep profiles private; think before you post; never share your ID number, address or holiday dates; use strong, different passwords and MFA; check app permissions; log out on shared computers. POPIA gives you rights over your personal information.

Location-based services (LBS) use the position of your device, found with GPS satellites, cell towers or Wi-Fi: navigation, ride-hailing, food delivery, local weather, “find my phone”. Benefits: convenience, safety and recovery of lost phones. Risks: your movements can be tracked, sold to advertisers or used for stalking, and criminals can learn when a house is empty. Allow location only while an app is in use.

LEARN IT

Careers, globalisation and Big Data

Web designer: plans the look and feel. **Web developer**: writes the HTML, CSS and code. **Multimedia designer**: creates graphics, animation, audio and video. **Database administrator**: secures, backs up and tunes databases. **SEO specialist**: helps a site rank higher in search results. **Cybersecurity analyst**: watches for attacks and protects systems.

Globalisation means people and countries are more closely linked by trade, travel and the internet. A crafts-person in Khayelitsha can sell overseas and families stay in touch across continents, but local shops can lose customers to giant online stores, some jobs move to cheaper countries and local cultures can be pushed aside. **Big Data and IoT** bring better healthcare and smarter cities, but also constant surveillance, profiling and data breaches, which is why laws such as POPIA matter.

3.2 Active or passive? [5]

Circle the kind of digital footprint that each item creates.

1	You post a photo on Instagram.	**Active**	**Passive**
2	A website records your IP address and the pages you visited.	**Active**	**Passive**
3	You fill in an online form to enter a competition.	**Active**	**Passive**
4	A free game app records your location in the background.	**Active**	**Passive**
5	Cookies track which products you looked at.	**Active**	**Passive**

3.3 Match the career [5]

Choose the career from the word bank for each task.
Word bank: cybersecurity analyst · database administrator · multimedia designer · SEO specialist · web designer · web developer

Task	Career
Draws a plan of the colours, fonts and layout of a new school website	
Writes the HTML and code that make the login page work	
Makes the animated intro and video for the home page	
Makes a daily backup of the school's learner database	
Notices that someone is guessing passwords and blocks them	

3.4 Protect your identity **[4]**

Explain **two** things a learner can do to protect their online identity. For each, say **why** it helps.

3.5 Discuss: location-based services **[4]**

Discuss **one benefit** and **one risk** of location-based services for a teenager, in a short paragraph.

3.6 Discuss: globalisation **[4]**

Discuss **one positive** and **one negative** social implication of globalisation made possible by the internet.

How did I do with this unit? ☐ I got it ☐ Nearly there ☐ I need help

One thing I want to ask my teacher:

UNIT 4 Revision workout: practical part 1

Arrays, searching, sorting, parallel arrays, strings and dates are things you have already done. This unit is a timed workout on paper: no computer, and every trace table filled in by hand, one step at a time.

By the end of this unit I can: (tick each box when you are sure)

- ☐ trace sorts and searches by hand
- ☐ use parallel arrays to find a name and a mark
- ☐ predict the result of string and date methods
- ☐ find the errors in array code
- ☐ write array and string code on paper

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

4.1 Sort the tuck-shop prices

Six cards show prices in rand: **14, 9, 22, 7, 18, 11**. Sort them using only this rule: *compare two neighbours and swap them if the left one is bigger*. Go left to right once to complete a pass. Write the row after **each pass** and stop after a pass with no swaps. Which card always ends at the far right after pass 1?

LEARN IT

Array patterns, searches and string methods

Array questions use a few patterns: a **running total** (start at 0), a **counter**, the **highest so far** (start with the first element, not 0) and a **search**. A **linear search** checks one item after another and works on any list; a **binary search** needs a **sorted** list and throws away half of it with every look. Use a Boolean flag, and move on only when the item was not found. **Parallel arrays** share an index: arrName[4] and arrMark[4] belong to the same learner. Never sort one array without swapping the other.

Method	What it does	Example
Pos(sub, s)	Position of the first match; 0 if none	Pos('p', 'Cape') = 3
Copy(s, start, count)	A piece of s	Copy('Cape Town', 6, 4) = 'Town'
Delete(s, start, count)	Removes a piece from s itself	Delete(s, 1, 5) drops the first 5 characters
Insert(sub, s, pos)	Puts sub into s itself	Insert('X', s, 1) puts X in front
EncodeDate(y, m, d), DaysBetween(later, earlier)	Builds a date; whole days between two dates (DateUtils)	EncodeDate(2026, 11, 3)
FormatDateTime(fmt, d), IncDay(d, n)	Date as text; n days later	'dd mmmm yyyy' gives 03 November 2026

4.2 Be the computer: bubble sort [6]

arrFare holds the taxi fares **29, 15, 36, 8, 20, 12**. Write the array after each pass of the outer loop and the number of swaps in the pass.

Delphi

```
for i := 1 to 5 do
  for j := 1 to 6 - i do
    if arrFare[j] > arrFare[j + 1] then
    begin
      iTemp := arrFare[j];
      arrFare[j] := arrFare[j + 1];
      arrFare[j + 1] := iTemp;
    end;
```

Pass (i)	Array after the pass	Swaps in the pass
1		
2		
3		
4		
5		

4.3 Be the computer: selection sort [5]

arrN holds **31, 12, 25, 8, 19**. For each pass write the position of the smallest item (iMin) and the array after the swap.

Delphi

```
for i := 1 to 4 do
begin
  iMin := i;
  for j := i + 1 to 5 do
    if arrN[j] < arrN[iMin] then
      iMin := j;
  if iMin <> i then
  begin
    iTemp := arrN[i];
    arrN[i] := arrN[iMin];
    arrN[iMin] := iTemp;
  end;
end;
```

Pass (i)	iMin after the inner loop	Array after the pass
1		
2		
3		
4		

4.4 Be the computer: binary search [5]

arrCode is sorted: 105, 118, 124, 137, 142, 151, 166, 172, 189, 195 (positions 1 to 10). The code searches for **166**.

Delphi

```
iLow := 1;  iHigh := 10;  bFound := False;
while (iLow <= iHigh) and (not bFound) do
begin
  iMid := (iLow + iHigh) div 2;
  if arrCode[iMid] = 166 then
    bFound := True
  else if arrCode[iMid] < 166 then
    iLow := iMid + 1
  else
    iHigh := iMid - 1;
end;
```

Round	iLow	iHigh	iMid	arrCode[iMid]
1				
2				
3				
4				

4.5 Be the computer: parallel arrays [5]

arrName holds Anele, Bongani, Chloe, Dumisani, Esihle, Fatima and **arrMark** holds 64, 71, 58, 85, 49, 77 (positions 1 to 6). Complete the loop table. Afterwards the learner at position iBest is the top learner: write the name here: _____.

Delphi

```
iTotal := 0;
iBest := 1;
for i := 1 to 6 do
begin
  iTotal := iTotal + arrMark[i];
  if arrMark[i] > arrMark[iBest] then
    iBest := i;
end;
```

i	arrMark[i]	iTotal	iBest
1	64		
2	71		
3	58		
4	85		
5	49		
6	77		

4.6 Strings and dates [7]

Let sText be 'load shedding'. Pos(' ', sText) = _____, Copy(sText, 1, 4) = _____ and Length(sText) = _____. After Delete(sText, 1, 5) the variable sText holds _____. DaysBetween(EncodeDate(2026, 11, 3), EncodeDate(2026, 10, 27)) = _____. FormatDateTime('dd mmmm yyyy', EncodeDate(2026, 11, 3)) gives _____, and DayOfTheWeek of that date is _____ (Monday = 1).

4.7 Debug: the class average [4]

This code should show the average of 30 marks and the **highest mark** (not its position). Find **four** errors.

```Delphi
iHigh := 0;
for i := 1 to 30 do
begin
  iTotal := iTotal + arrMark[i];
  if arrMark[i] > iHigh then
    iHigh := i;
end;
rAvg := iTotal / 31;
lblOut.Caption := 'Average: ' + rAvg + ' High: ' + IntToStr(iHigh);
```

4.8 Write it: find a learner [6]

The arrays **arrName** and **arrMark** hold **iCount** learners. The name to find is in **edtName**. Use a Boolean flag to search **linearly** and show the name and mark in **lblOut**, or *Not found*. Ignore capital letters.

4.9 Write it: is this ID valid? **[7]**

The ID number is in **sID**. Set **bValid** to True only if it has **13 characters**, every character is a **digit** and the month (characters 3 and 4) is from 01 to 12. Show *Valid* or *Invalid* in **lblOut**.

How did I do with this unit? ☐ I got it ☐ Nearly there ☐ I need help

One thing I want to ask my teacher:

UNIT 5 Revision workout: practical part 2

Text files, procedures, functions and database tables are the three tools for the PAT. They look different, but each uses the same loop shape. Work through this unit with a pencil only.

By the end of this unit I can: (tick each box when you are sure)

- ☐ trace a loop that reads a file and splits each line
- ☐ trace procedures and functions, including VAR parameters
- ☐ trace a loop over the records of a table
- ☐ find the errors in database code
- ☐ write code to read a file and add a record

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

5.1 Photocopy or original?

Mrs Naidoo hands a learner a **photocopy** of her mark sheet; the learner writes 100 next to every name on the copy. Another learner gets the **original** sheet and does the same. What does Mrs Naidoo's sheet look like afterwards in each case? Which is like a **value** parameter and which like a **VAR** parameter?

THINK FIRST · no computer needed

5.2 A finger on the paper table

Here is a table of learners:

ID	Name	Grade	Mark
1	Anele	10	55
2	Bongi	11	72
3	Chloe	11	48
4	Dumi	12	81
5	Esihle	11	66
6	Fatima	10	90

Put a finger on the first row. Write the steps, in plain English, that count the learners in Grade 11. What do you do when your finger moves off the bottom? What happens if you forget to move your finger down?

LEARN IT

One pattern in three costumes

Arrays, text files and tables all use one shape: **start, keep going while there is more, deal with one item, move on**.

	Array	Text file	Database table
Start	i := 1	Reset(tFile)	tblLearner.First
Keep going while	i <= iCount	not EOF(tFile)	not tblLearner.Eof
Get the item	arrMark[i]	ReadLn(tFile, sLine)	tblLearner['Mark']
Move on	Inc(i)	ReadLn does it	tblLearner.Next

Edit then **Post** changes a record; **Insert** then **Post** adds one; **Delete** removes it and Delphi moves to the next record by itself, so call Next only when nothing was deleted. A **procedure** does a job; a **function** returns one value through **Result**, for example *Result := rPrice * iQty*. A **value** parameter is a copy; a **VAR** parameter is the caller's own variable, so changes stay.

Delphi

```
tblLearner.First;                // delete every Grade 12 record
while not tblLearner.Eof do
  if tblLearner['Grade'] = 12 then
    tblLearner.Delete
  else
    tblLearner.Next;

procedure AddBonus(var iMark: Integer; iBonus: Integer);
begin
  iMark := iMark + iBonus;
  if iMark > 100 then iMark := 100;
end;                              // AddBonus(iScore, 5) changes iScore
```

5.3 Predict: value and VAR [3]

Write exactly what **lblOut** shows. Trace each call.

Delphi

```
procedure Change(iA: Integer; var iB: Integer);
begin
  iA := iA * 2;
  iB := iB + iA;
end;

// in the button code
iX := 5;  iY := 10;
Change(iX, iY);
Change(iY, iX);
lblOut.Caption := IntToStr(iX) + ' ' + IntToStr(iY);
```

5.4 Be the computer: read and split [6]

Marks.txt contains four lines: **Thabo,56**, **Aisha,48**, **Sipho,71**, **Lindiwe,50**. Complete a row for every check of the loop; the last row is the check that ends it.

Delphi

```
iPass := 0;  iBest := 0;  sBest := '';
AssignFile(tFile, 'Marks.txt');
Reset(tFile);
while not EOF(tFile) do
begin
  ReadLn(tFile, sLine);
  iPos := Pos(',', sLine);
  sName := Copy(sLine, 1, iPos - 1);
  iMark := StrToInt(Copy(sLine, iPos + 1, Length(sLine)));
  if iMark >= 50 then
    Inc(iPass);
  if iMark > iBest then
  begin
    iBest := iMark;
    sBest := sName;
  end;
end;
CloseFile(tFile);
```

Check	EOF(tFile)	sName	iMark	iPass	sBest
1	False				
2	False				
3	False				
4	False				
5	True				

5.5 Be the computer: a database loop [6]

tblLearner is the table in Think First (IDs 1 to 6). Complete a row for each record the loop visits.

Delphi

```
iCount := 0;
iTotal := 0;
tblLearner.First;
while not tblLearner.Eof do
begin
  if tblLearner['Grade'] = 11 then
  begin
    Inc(iCount);
    iTotal := iTotal + tblLearner['Mark'];
  end;
  tblLearner.Next;
end;
```

Record (ID)	Grade	Grade = 11 ?	iCount	iTotal
1	10			
2	11			
3	11			
4	12			
5	11			
6	10			
past the last record	-			

5.6 Debug: bonus marks [4]

This code should give **5 bonus marks** to every Grade 10 learner, but not above 100. Find **four** errors.

Delphi

```
tblLearner.First;
while not tblLearner.Eof do
begin
  if tblLearner['Grade'] = '10' then
  begin
    tblLearner['Mark'] := tblLearner['Mark'] + 5;
    if tblLearner['Mark'] > 100 then
      tblLearner['Mark'] := 100;
  end;
end;
```

5.7 Write it: file into arrays [8]

Results.txt has one line per learner in the form *Thabo,56*. Read the whole file into **arrName** and **arrMark** (array[1..100]), count the learners in **iCount** and show the average mark to one decimal in **lblAvg**. (tFile, sLine, iPos and iTotal are declared.)

5.8 Write it: add a record [6]

The name is in **edtName**, the grade in **sedGrade** (a spin edit) and the mark in **edtMark**. Add a new record to **tblLearner**, but only if the name is not empty and the mark is from 0 to 100. Otherwise show a message.

How did I do with this unit? ☐ I got it ☐ Nearly there ☐ I need help

One thing I want to ask my teacher:

UNIT 6 Revision workout: theory on paper

Paper 2 tests whether you can explain how computers, networks and databases work and why things go wrong. This workout covers hardware, the operating system, networks, security, databases and data quality, with short matching tasks and written answers.

By the end of this unit I can: (tick each box when you are sure)

- ☐ describe the key hardware parts and what the operating system does
- ☐ match protocols and security threats to their jobs
- ☐ use database terms correctly and draw a simple ERD
- ☐ choose the right validation check for a field
- ☐ explain data quality and security in a paragraph

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

6.1 The slow lab

The school lab has ten-year-old PCs. Each takes four minutes to start, they freeze when three programs are open, the hard-drive light is always on and pop-up adverts keep appearing. Suggest **three different causes** and say what you would do about each. Write full sentences.

THINK FIRST · no computer needed

6.2 The messy class list

A class list is kept in one table:

Learner	Class	Class teacher	Room
Lerato Mokoena	11A	Mr Naidoo	Room 12
Sam Adams	11A	Mr Naidoo	Room 12
Thabiso Khumalo	11A	Mr Naidoo	Room 12
Anke Botha	11B	Mrs Jacobs	Room 8
Pule Sithole	11B	Mrs Jacobs	Room 8

Mr Naidoo's class moves to Room 15. How many rows must change? What goes wrong if someone changes only two? How could **two tables** store the room only once?

LEARN IT

Revision map

Topic	What to remember
Hardware and OS	**CPU** runs instructions; **cache** is tiny, very fast memory beside it. **RAM** is volatile working memory; **ROM** permanently holds the BIOS, which runs the **POST** at start-up. An **SSD** (flash, no moving parts) is faster than a **HDD** but costs more per GB. The **operating system** manages memory, files, devices (via drivers), users and security, and provides the interface.
Networks	In a **star** network one cable fault affects one PC, but a switch fault stops everyone. **SMTP** sends e-mail, **POP3** downloads it, **IMAP** keeps it on the server; **FTP** moves large files; **HTTP/HTTPS** carry web pages; **VoIP** carries calls.
Security	Malware: virus (needs a file), worm (spreads alone), trojan (disguised), rootkit (hidden), ransomware, spyware, adware. Attacks: phishing, pharming, spoofing, SQL injection. Remedies: antivirus, firewall, strong passwords, access rights, encryption, updates, MFA, UPS and the 3-2-1 backup rule.
Databases and data quality	A **primary key** is unique; a **foreign key** holds another table's primary key. Relationships: 1:1, 1:M, and M:M (solved with a link table). **Referential integrity**: a foreign key must match an existing primary key. **GIGO**: wrong input gives wrong output. **Validation** (by the program) checks that data is reasonable; **verification** checks that it matches its source.

6.3 Computer systems and networks [8]

Write the term from the word bank next to each description.
Word bank: adware · cache · FTP · pharming · POP3 · POST · ransomware · SSD

Description	Term
Tiny, very fast memory next to the CPU that holds recently used instructions	
The check of RAM, keyboard and storage when the computer is switched on	
Storage with no moving parts, using flash memory	
The protocol that downloads e-mails to one device	
The protocol used to upload a large file to a server	
Sends you to a fake website even though you typed the right address	
Encrypts your files and demands payment to unlock them	
Shows pop-up adverts that you did not ask for	

6.4 Databases and data quality [8]

Write the term from the word bank next to each description.
Word bank: field · foreign key · length check · presence check · primary key · range check · record · referential integrity

Description	Term
One column in a table, such as Surname	
One complete row, holding everything about one learner	
A field whose value is unique for every record	
A field that holds the primary key of another table	
The rule that every foreign key must match an existing primary key	
A mark must be from 0 to 100	
A cell number must have exactly 10 digits	
The surname box may not be left empty	

6.5 Draw the ERD [5]

A library member can borrow **many books** over time, and each book can be borrowed by **many members**. Draw an ERD with **tblMember**, **tblBook** and a link table **tblLoan**. Write the fields, mark the primary keys (PK) and foreign keys (FK), and label each line 1 or M at the correct end.

6.6 Explain: referential integrity [3]

tblLearner has a foreign key **ClassID** that refers to tblClass. Explain what **referential integrity** prevents here.

6.7 **Explain: GIGO** **[3]**

A clerk types the mark 98 as 89. Explain what **GIGO** means using this example, say why a range check of 0 to 100 would **not** catch it, and name a method that would.

6.8 **Describe: ransomware** **[4]**

Describe what happens to a school if **ransomware** infects the office PC. Explain **one** way to prevent the attack and **one** way to recover without paying.

How did I do with this unit? ☐ I got it ☐ Nearly there ☐ I need help

One thing I want to ask my teacher:

UNIT 7 PAT checklist and exam skills

Two things decide your final marks: the quality of your PAT and the way you handle the exam paper. This unit gives you a one-page audit for your PAT and the exam habits that protect marks.

By the end of this unit I can: (tick each box when you are sure)

- ☐ audit my PAT against a marker's checklist
- ☐ decide how much to write from the mark allocation and command word
- ☐ turn bullet points into a paragraph with a point and a reason
- ☐ plan code in comments and show my logic
- ☐ work out how many minutes I have for each mark

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

7.1 Which answer would you mark?

Explain why a school should keep a copy of its data at a second place. [3]

Answer A: "– fire – theft – flood"

Answer B: "If all the data is in one building, a fire, flood or burglary can destroy the computer and its backup together. A copy at a second place, such as the cloud, survives such a disaster. The school can then restore its records and carry on."

Which answer shows understanding? Why can a marker not give full marks to A?

LEARN IT

Read, count, plan, answer

Read the whole question and underline the command word. **Count** the marks: each mark is usually one correct, separate point, so [3] means three points. **Plan**: jot the points in the margin. **Answer** in sentences with *because*, *so* and *for example*, so each point has its reason; a marker may stop crediting a list after the first few bullets but reads a paragraph in full. **Check** that you gave as many points as there are marks. In code questions **logic earns marks before syntax**: write your plan as comments first, then the code under each comment; if time runs out the comments still show your thinking. In traces fill every row, because the method earns marks.

Command word	What to do
Name / State / List	Give the answer only: a word or short phrase for each mark.
Explain	Say what and why or how: a point and its reason for each mark.
Discuss	Give both sides (advantages and disadvantages) and explain each, in a paragraph.
Differentiate	Show how two things differ, with both things in every sentence.
Motivate / Justify	Choose, then give explained reasons.
Calculate	Formula, working and answer with a unit.

7.2 My PAT audit

Rate each row **0** (not done), **1** (partly) or **2** (fully) and write the evidence or what you still need to fix before the hand-in date.

Area	What a marker looks for	Rate	Evidence or what I must fix
Planning	Clear description of the problem and users; data planned (fields, types, keys, ERD); a sketch of every form		
Code layout	Header comment, indentation, Hungarian prefixes, comments on each method		
Methods	Procedures and functions with parameters, each doing one job; no repeated code		
Arrays and loops	Used to search, sort or total data in a meaningful way		
Text file	AssignFile, Reset/Rewrite/Append, EOF, CloseFile; FileExists checked; no LoadFromFile or SaveToFile		
Database	Related tables; records added, edited and deleted in code; searching with a loop; no SQL		
Validation	Inputs checked (presence, range, type); helpful messages; the program never crashes		
User interface	Clear labels, logical tab order, consistent look, buttons enabled only when sensible		
Testing	Test table with normal, boundary and invalid data; expected and actual results		
Documentation	User guide with screenshots; technical notes (ERD, list of methods)		
Hand-in	Runs on another PC (relative paths, database in the project folder); folder backed up; declaration signed; on time		

7.3 Bullets to a paragraph [3]

A learner answered *Explain how phishing works. [3]* with: "– fake e-mail – click link – steals password". Rewrite it as a paragraph of 3 or 4 sentences with a reason in each.

7.4 How long per mark? [4]

A paper has 75 marks and lasts 120 minutes. That gives _____ minutes per mark, so a 6-mark question should take about _____ minutes. If you keep _____ minutes at the end to check, you have _____ minutes for the questions, which is about _____ minutes per mark.

7.5 Plan first, then code **[6]**

A pie costs R14. The number of pies is in **edtPies** and the money handed over in **edtPaid**. Show the change in **lblOut**, or *Not enough money*. (a) Write your plan as comments. (b) Write the code under your comments, with variable declarations.

How did I do with this unit? ☐ I got it ☐ Nearly there ☐ I need help

One thing I want to ask my teacher:

TERM CHECKPOINT

Term 4 checkpoint: the connected world and revision

Time: 120 minutes	**Total:** 73 marks	**Mark:** _____ / 73

No notes, no computer. Show all your working. Write Delphi code neatly and use correct indentation.

SECTION A — Trace and predict [12]

A1 [4]

arrFare holds the taxi fares **24, 18, 30, 12, 21**. Write the array after each pass and the number of swaps in the pass.

Delphi

```
for i := 1 to 4 do
  for j := 1 to 5 - i do
    if arrFare[j] > arrFare[j + 1] then
    begin
      iTemp := arrFare[j];
      arrFare[j] := arrFare[j + 1];
      arrFare[j + 1] := iTemp;
    end;
```

Pass (i)	Array after the pass	Swaps in the pass
1		
2		
3		
4		

A2 [3]

What does **lblOut** show? Trace each call carefully.

Delphi

```
procedure Mix(var iA: Integer; iB: Integer);
begin
  iA := iA + iB;
  iB := iA - iB;
end;
iP := 4;  iQ := 7;
Mix(iP, iQ);
Mix(iQ, iP);
lblOut.Caption := IntToStr(iP) + ' and ' + IntToStr(iQ);
```

A3 **[5]**

Scores.txt contains three lines: **Zodwa,72**, **Kabelo,38** and **Imran,64**. Complete a row for each check of the loop; the last row is the check that ends it.

Delphi

```
iTotal := 0;  iFail := 0;  iLow := 100;
AssignFile(tFile, 'Scores.txt');
Reset(tFile);
while not EOF(tFile) do
begin
  ReadLn(tFile, sLine);
  iP := Pos(',', sLine);
  iMark := StrToInt(Copy(sLine, iP + 1, 3));
  iTotal := iTotal + iMark;
  if iMark < 50 then
    Inc(iFail);
  if iMark < iLow then
    iLow := iMark;
end;
CloseFile(tFile);
```

Check	EOF(tFile)	iMark	iTotal	iFail	iLow
1	False				
2	False				
3	False				
4	True				

SECTION B — Find the errors [7]

B1 [4]

This code should show the mark of the learner whose name is in **edtName**. Find and correct **four** errors.

Delphi

```
bFound := True;
i := 1;
while (i < iCount) or (not bFound) do
begin
  if arrName[i] = edtName.Text then
    bFound := True
  else
    Inc(i);
end;
if bFound = True then
  lblMark.Caption := arrMark[i]
else
  lblMark.Caption := 'Not found';
```

B2 [3]

This code should remove every member whose fines are more than R500, but it sometimes misses a member. Explain what is wrong and write the corrected loop.

Delphi

```
tblMember.First;
while not tblMember.Eof do
begin
  if tblMember['Fines'] > 500 then
    tblMember.Delete;
  tblMember.Next;
end;
```

SECTION C — Write the code [24]

C1 [6]

The parallel arrays **arrPlayer** and **arrScore** (array[1..20]) hold **iCount** players. Write a function **FindTop** that returns the **position** of the highest score. Then write the lines that show the top player's name and score in **lblBest**.

C2 [8]

Players.txt has one line per player in the form *Zodwa,72*. Copy only the players with a score of **70 or more** to a new file **TopPlayers.txt**, each as *Zodwa - 72*, and show in **lblCount** how many were copied. (File variables tIn and tOut; sLine, iPos, sName, iScore and iCount are declared.)

C3 **[6]**

tblStock has the fields **ItemName** and **Qty**. Visit every record. If Qty is less than 5, add 20 to Qty, save the change and add the ItemName to **memReorder**. At the end show in **lblNum** how many items were topped up.

C4 **[4]**

Write a procedure **MinMax** with integer parameters **iA**, **iB**, **iC** and **VAR** parameters **iSmall** and **iBig** that return the smallest and largest of the three. Then write the call using **edtA**, **edtB**, **edtC** and the variables iLow and iHigh.

SECTION D — Theory [30]

D1 [4]

Circle the V of Big Data that each situation shows best.

1	Sensors in a mine send new gas readings every second that must be analysed at once.	**Volume** **Velocity** **Variety**
2	A streaming company stores millions of hours of film.	**Volume** **Velocity** **Variety**
3	A bank combines customer tables, scanned IDs, call recordings and social media comments.	**Volume** **Velocity** **Variety**
4	Hospital heart monitors produce new readings every second that doctors need immediately.	**Volume** **Velocity** **Variety**

D2 [5]

A photo is **1 600 × 900 pixels** at **3 bytes** per pixel (1 KB = 1 024 bytes; 1 MB = 1 024 KB).

Question	Answer
(a) Number of pixels	
(b) File size in bytes	
(c) File size in MB, to 2 decimal places	
(d) Size in bytes after JPEG compression at 12:1	
(e) Is JPEG lossy or lossless?	

D3 [4]

A game farm fits IoT sensors to its fences and water troughs. **Discuss** one benefit and one risk of this system.

D4 **[3]**

Differentiate between HTTP and HTTPS, and say why the difference matters when you pay online.

D5 **[3]**

Explain how an OTP or a security token makes logging in safer than a password alone.

D6 **[4]**

Differentiate between an active and a passive digital footprint with an example of each, and explain one way a learner can limit their footprint.

D7 **[4]**

Write the validation check from the word bank.
Word bank: length check · list check · presence check · range check · type check

Rule	Check
A cell number must have exactly 10 digits	
A mark must be from 0 to 100	
The name box may not be empty	
The province must be chosen from nine given names	

D8 **[3]**

Explain the purpose of a **foreign key**, using tblLearner and tblClass as your example.

COMING UP NEXT TERM · warm-up

Asking questions of a table

In Grade 12 you will ask a database questions such as "show me only the Grade 12 learners". You do not need code to think like this: only a table, a pencil and clear questions. Here is a class table:

Surname	Name	Grade	Mark	Sport
Dlamini	Thabo	12	68	Soccer
Naidoo	Priya	11	74	Netball
Botha	Kobus	12	59	Rugby
Khumalo	Zanele	12	81	Netball
Pieterse	Liam	11	45	Soccer
Mokoena	Lerato	12	72	Netball
Adams	Cleo	10	90	Hockey
Jacobs	Imran	10	63	Soccer
Sithole	Pule	11	58	Rugby
Van Wyk	Anke	11	77	Hockey

Answer in plain English. Write the rows that you would show.

(a) **Show only learners in Grade 12, sorted by surname.**

(b) **How many** learners play netball?

(c) What is the **average mark** of the Grade 12 learners?

(d) Show the learners with a mark **above 70**, highest mark first.

(e) How would you find out **how many learners play each sport**? Describe the steps.